
What are TLS & SSL?

Both **TLS (Transport Layer Security)** and **SSL (Secure Sockets Layer)** are cryptographic protocols that **secure communication over the internet** 🌐.

Quick Note:

- **SSL** is the **older protocol**, now considered **deprecated** ❌.
- **TLS** is the **modern, secure replacement** ✅.
- But people still often say “SSL certificate” even when they mean **TLS certificate** 😊.

What is a TLS/SSL Certificate?

A **TLS/SSL Certificate** is a **digital file** issued by a **Certificate Authority (CA)** that:

- ✅ Proves the **identity** of a website (authentication)
- 🗝️ Enables **encrypted communication** between a client (e.g., browser) and a server

How TLS/SSL Works – Step by Step

1. Client Hello

Browser says:

"Hi, I want to connect securely! Here are the protocols and ciphers I support."

2. Server Hello

Server replies:

"Cool! Here's my **TLS certificate** (with my public key). Let's use these settings."

3. Certificate Validation

Browser checks:

- Is this cert issued by a **trusted CA**? 
- Is it **expired**? 
- Is the **domain name correct**? 

If valid → continue 

If not →  Security warning

4. Key Exchange

They use **asymmetric encryption** (public/private keys) to securely agree on a **shared secret key**.

5. Encrypted Communication

From now on, all communication is **encrypted using symmetric encryption** (faster).

What Info Does a TLS/SSL Certificate Contain?

- ♦ Domain name (e.g., www.example.com)
- ♦ Organization info
- ♦ Public key
- ♦ Expiry date

- ♦ Issuer (CA)
- ♦ Signature (proof it's legit)

Types of TLS/SSL Certificates

Type	Use Case	Features
 DV (Domain Validation)	Basic websites	Validates domain only
 OV (Organization Validation)	Business sites	Validates org identity
 EV (Extended Validation)	Financial sites	Strictest validation (shows green bar/padlock)

Why Use TLS/SSL?

-  Data protection (e.g., login info, credit cards)
-  PCI DSS compliance for e-commerce
-  SEO boost (Google favors HTTPS)
-  Trust (padlock  icon in browser)

Real-Life Example

When you go to <https://google.com>:

- **https** means it uses **TLS**
 - The  icon shows the **certificate is valid**
 - Your search data is encrypted end-to-end 
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